

# Income constraints on Mobility - Social Capital under Inequality

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## Abstract

The effects of income inequality on labour motivation have generated an immense debate at a time when mobility has remained stagnant. While argued as a key determinant in the discourse on inequality, status considerations are rarely endogenised in economic models and limit the understanding of long-term interaction of status with inequality. To explain status motivations under long-term inequality, we treat opportunities for mobility as social capital accumulations and develop a framework to examine how income inequalities may affect status motivation through differential access to social capital. The effect of relative income is thus explored in an economy with a dynamic hierarchy of rich and poor positions that participants are able to move between. With the participants optimising a wealth utility with diminishing returns, the equilibrium conditions indicate that a rise in income inequality does provide more incentive to participate in status competitions, but such competitions become infeasible when inequality rises too much. Further, with participants bearing costs for mobility, the equilibrium conditions suggest only two states in the long-run - one where both rich and poor strive for status and the other where only the rich participate in status competitions. The participation costs also play a significant role in the competitions - suggesting that reduction of such costs may constitute an effective policy tool to mitigate social disparities rooted in income inequality.

## 1 Introduction

The consumer response to relative income has been of interest to economists for at least as far back as Veblen (1899) - but the precise mechanisms through which economic inequality shapes consumption and labor participation remain contested. In recent times, some perspectives have treated inequality as a necessary incentive structure that motivates economic effort (Baron, 2017) - while others argue that inequality suppresses motivation (Gesiarz, De Neve, & Sharot, 2020). The motivation to participate is often influenced by their perceived social status or prestige - a construct that has posed measurement challenges in both economics and sociology (Wegener, 1992).

The current paper focuses on two particular issues with the assessments of status motivations. First, it emphasises the need to endogenise status within formal economic models. Despite a growing literature that has explored status formation through linguistic communication and competitions (Boltanski & Thévenot, 1983; Ridgeway, 1991; Gould, 2002; Leicht, 2022), status has more often been understood in economics as a set of static attributes. The conventional thinking (Hurst, Gibbon, & Nurse, 2016) treats status qualitatively - operationalising low status via categorical group membership (race, gender, ethnicity, region etc.) - an approach that imposes limitations on quantitative assessments of status. Further, as status perceptions change in response to economic growth and inequality, a failure to endogenise status in an economic model obscures long-term effects of inequality on status motivations.

Second, it distinguishes *subjective* status – the individual’s internalized sense of social position from the *objective* status – positions with consequences for economic opportunity.

While consumer focus on subjective status may have increased in the recent environment of stagnated economic mobility (Frank, 1993), it cannot be ruled out that status differences caused by long-term inequalities continue to set limits on labour or consumption towards objective status (Hirsch, 1977). A distinction between objective and subjective aspects of status helps us better delineate how inequality may influence particular mechanisms of status formation.

Building on the above clarifications, the paper focuses on a particular mechanism of how inequality affects participation in the long-term - examining how objective status is subjected to social capital investments. Rather than being ascribed by static social groupings, status is viewed as a dynamic state changed through expenditures aimed at maintaining one's position within privileged networks. The long-term equilibrium conditions of the model - that we further detail - reveal three main observations on how inequality affects social capital investments. First, it becomes clear that while inequality can initially encourage social capital investment, high levels of inequality impose constraints that make status competitions unsustainable - thus reconciling Frank's and Hirsch's seemingly opposing - as cultural adaptation can encourage individuals to meet local status pressures (subjective status) despite a decline in structural mobility (objective status). Second, high participation costs, driven by social closure, are crucial in maintaining status hierarchies and may serve as a necessity for status competitions. Third, the results permit only two stable equilibria: one where both rich and poor compete for status, and another where only the rich participate.

The objective sense of status serves as a link between inequality and social capital investments which respond to objective status motivations. Status only has an indirect (instrumental) effect on consumer wealth in the model (Cole, Mailath, & Postlewaite, 1992; Postlewaite, 1998). Drawing on recent sociological approaches, the paper considers power merely an unequal access to opportunity (Tilly, 1998; Kendall, 2002) - rather than as hierarchical deference (Coleman, 1990). Taking an approach that follows Goode (1978) closely, the paper argues that objective status may stem from both social capital (e.g., networks) and inherent traits (e.g., ability). The opportunities for success are represented in the model as probability functions for promotion to a higher status position - that are in turn subject to social capital investments. In other words, the participants compete both with raw talent and acquired skills (social capital) to get promoted to (or to maintain) a "rich" position while having different budgets for social capital investments. The winner of such a game is promoted to a rich position (or maintains her rich position) by having the maximum total score comprising both of raw-talent  $r_i$  (uniformly distributed in a population) and social capital  $s_i$  (enhanced with expenditure).

The long-term dynamics of the model are explored with two types of participants - the poor and the rich - a theoretical simplification aligned with the conflict-based theories of status contests (Weber, 1925; Tilly, 1998). The rich participants in the game benefit with a higher disposable income ( $y_2$ ) solely on grounds of their rich status while the poor participants receive a disposable income  $y_1 < y_2$ . It is worth emphasising that those who don't participate in the tournament face a loss of status, but those who do spend only benefit probabilistically. A useful analogy to explain such a competition is that of tennis leagues or golf tournaments - where rewards are risky but are unlike lotteries (i.e. more expenditure does not linearly increase one's chance to win). As with tennis or golf, participants with appropriate raw-talent and skills may exist naturally in a population but their success also depends on institutional engagement with other players. More specifically, there may exist biological characteristics such as strength, dexterity, height or psychological attributes (resilience, composure etc.) that help one's success, but there also exist several characteristics such as technique, rankings, knowledge of equipment etc. that are only available at a significant cost to a participant. The enhancement of these latter skills (social capital) are what the expenditure-depending institutional support are meant to represent in the above probabilistic game. Several real-life competitions for objective status are similar to such tournaments where interaction is small-scale as well as personal (Goode, 1978) - with social capital investments playing a significant role (e.g. in interviews, formal

introductions and business interactions).

Rather than encompassing the entire population, the competitions thus considered are small-scale tournaments that employ varying success criteria across several stages. The competitions are thus meant to be independent and vary only in the threshold participation cost and range of distribution of  $r_i, s_i$  of the participants. The presented model considers a single instance of a competition with known distribution of  $r_i$  and  $s_i$  among a relatively small number of participants who exercise their choice to spend on social capital without an *a priori* knowledge of  $r_i$  and  $s_i$ . In a limited context of social capital investments, the results of such a model have implications on how inequality shapes investment decisions made with a perception of mobility and how such concerns may aggregate into macroeconomic trends.

A closed-form solution with a diminishing utility from wealth and a threshold level expenditure required for participants to spend (in order to avail social capital) suggests that there are only states in the long-term equilibrium. First, where the rich as well as the poor participant participates in the competitions for objective status. Second, where only the rich participant engages in such competitions. A state where only the poor participate is ruled out in the above formulation. Thus if the social capital investments were to be seen as an exchange between rich and the poor (Galbraith, 1958; Hirsch, 1977), the competitions where poor strive for status must ultimately become infeasible (except when relying thinly on a reference-dependent utility) and lead to only poor withdrawing from competitions for objective status. Despite concerns for rise in status goods, the normative argument from Hirsch about the limits to positional efforts still holds. It also becomes evident that participation costs - along with higher income differences - have a key role to play in the rise or decline in social capital investments (e.g. education or network-building activities). More particularly, social capital investments are likely to be subject to a discouragement through high participation costs while still being incentivised in an environment of higher income differences.

The above conclusions may also be relevant to the status consumption literature – which views expenditures on status goods as wasteful or as means to signal high status. The paper argues that many status-related expenditures – such as higher education, neighborhood choice, or professional affiliation – may serve a role of enabling access to upward mobility – emphasising that such social capital goods are not a mere symbol of status but a mechanism for acquiring it. This social capital perspective could thus clarify a slight ambiguity in the literature: whether rising inequality leads to an increase or decrease in status-seeking behavior. While the conventional view argues that a higher demand for positional goods had been a result of relative income equality in the post-war period (Hirsch, 1977), modern accounts finds that higher inequality fuels status consumption (Frank, 1993). The contrast against objective status in the model helps explain that it is possible for status-anxieties (responding to subjective status) to increase while the normative argument about limitations imposed by inequality on positional consumption (Hirsch, 1977) continues to hold in the context of objective status. While subjective status captures internal perceptions and psychological responses, objective status relates to an individual's actual structural position and their capacity to translate status investments into material returns.

In what follows, Section 2 surveys background literature pertaining to effects of inequality on economic participation and discusses the need to reconcile subjective and objective aspects of status. Section 3 describes the model and results. The implications of the model for social capital investments under inequality are then discussed in Section 4.

## 2 Literature Survey

The relationship between inequality, motivation, and status-seeking behavior has generated extensive debate across both economics and sociology. While early economic thought often emphasized the functional role of inequality as a motivational force—providing incentives for

effort, investment, and innovation—more recent perspectives have challenged this view, particularly in the context of constrained social mobility and stratified opportunity structures. Arguments in favor of inequality (Baron, 2017) often emphasize efficiency and incentive compatibility (Becker & Tomes, 1979), based on the idea that greater rewards reflect greater effort. However, competing empirical and behavioral research suggests that beyond a certain threshold, inequality can undermine labor participation, erode social trust, and reduce long-term productivity (Perotti, 1993; Alesina & Perotti, 1996; Stiglitz, 2012). These contrasting perspectives remain visible in recent debates, reflecting an unresolved tension: whether inequality spurs economic participation or discourages it by limiting realistic pathways to advancement (Chetty, Hendren, Kline, & Saez, 2014).

While status or prestige is often argued as a key motivator for labour participation as well as consumption (Bénabou & Tirole, 2003; Frank, 1993), status is often contrasted as a social goal against economic goals of a consumer – a dichotomy that complicates the endogenisation of status within economic models. At least since Weber (1925), a distinction is drawn between status and class – wherein status refers to fixed qualitative attributes such as race, regional identity, or gender, while class denotes one's economic position (Hurst et al., 2016). But while contrasting social closure against the idea of unified economic classes, Weber (1925) does draw attention to the exclusionary boundaries constructed by social groups (social closure) as well as the economic consequences brought about by such boundaries. The idea that status is orthogonal to economic goals – or the “pure status” view – seldom fits with the real-world where status may contribute to income or change in response to income (Moldovanu, Sela, & Shi, 2007). Indeed the other extreme view that economic goals alone motivate competitions for status – so that it is derived purely from economic rewards also remains unrealistic. As Moldovanu et al. (2007) argue, the reality is often somewhere in between the two extremes. Truys (2010) argue that without the interdependence of status and consumption/labour, status would be of little importance. The understanding of how economic inequalities translate into social hierarchies and status-based boundaries remains an area of active research in sociology and economics (Wegener, 1992; Bénabou & Tirole, 2003; Truys, 2010).

The theories for status in sociology have so far coalesced around two principal frameworks: *conflict*-based approaches, which emphasise power struggles and social closure, and *order*-based approaches, which focus on hierarchical positions e.g., occupations (Wegener, 1992). While *order*-based approaches (Coleman, 1990; Goldthorpe, 2000) have been relied upon more often in empirical studies (e.g., the EGP classes or the Goldthorpe class scheme), the challenge for both *conflict*-based and *order*-based approaches (Parsons, 1963; Blau, 1974) has been the reconciliation of subjective and objective aspects of status (Wegener, 1992). To recall, objective status is the observed reality of status differences while subjective status circumscribes the effect of status differences on individual behaviour. Empirical issues with measurement of status are likely if two two aspects aren't reconciled. For example, the status measured in occupational surveys can be at odds with the self-reported perceptions of status (Wegener, 1992).

The paper uses the notion of objective status to define the scope of the effects of income inequality on consumer motivations to participate. The competition-based view of status – drawing from Goode (1978) – emphasises how institutions discourage or promote the success-determining attributes in a society. Goode's framework also helps a departure from the widely criticized view of status as a tradable-power (Coleman, 1990) - as it rejects the notion of there being a fixed total amount of status in society (Wegener, 1992). While status influences economic outcomes, it is not viewed in the model as a wealth-determined goal. Instead, using an approach proposed by Postlewaite (1998), status is considered only of an indirect role in maximisation of wealth utility – so that the rank of the participant (rich or poor status) does not directly contribute to utility and matters only through a higher income that contributes to maximisation of the participant's expected wealth utility.

The presented approach also offers a circumspect view on the visibility-based models in

status consumption literature. When status is treated as fixed and not directly measurable, empirical approaches may often reduce status consumption to a form of futile expenditure. Such a framing, however, may be at odds with more functionalist perspectives (e.g. from Weber (1925); Goode (1978)) where status-seeking serves social purposes. For status goods that simultaneously enhance social capital, the effects of inequality may diverge from the predictions of visibility-driven models. Misinterpretations of status pressures – e.g. overreaching conclusions about status-seeking behaviour by the poor (Nieftagodien, Van Der Berg, et al., 2007; Burger, Louw, Pegado, & van der Berg, 2015) – are also possible if one fails to highlight that status consumption only reflects the subjective aspects of status.

The implications of this paper are situated primarily within the domain of long-term implications of objective status—where status motivations are observable through investments in social capital. Nonetheless, more detailed views of subjective of status provide a crucial dimension to research in economic participation (J. J. B. Mijs, 2019) . When examining status—whether from a normative or descriptive perspective—it is essential that the conclusions regarding its implications reconcile both its subjective and objective aspects.

### 3 Model

In the status game that selects the winner to be maintained at the rich position, only two characteristics matter for every individual. The first is the social capital that can be enhanced with expenditure by the individual participant and the second is a raw-talent that cannot be changed with higher expenditure. Both skill  $s_i$  (enhanced with expenditure) and the raw talent  $r_i$  (which cannot be changed with higher expenditure) are assumed to be uniformly distributed in the population. Assuming the weight  $\mu$  of raw-talent in the maximum total score considered for the win in a competition, consider the preferences for the probability of  $\mu r_i + (1 - \mu)s_i$  (where  $r_i$  is raw-talent and  $s_i$  are skills that are available with higher expenditure) under the assumption that the population remains constant over time (there are the same number of rich and poor in total) and that the participants are not aware of their  $r_i$ ,  $s_i$  before participating in the competitions. In other words, the rich participants are replaced by either a rich or poor participant only through the tournament that selects highest combined score regardless of rich or poor background of the participant. Since all participants can participate in every such tournament, it is possible for a rich participant to lose her rich-position and let another participant acquire her position.

It is assumed that the preferences for all poor participants are the same as are the preferences of all the rich participants. Having a rich or a poor status by itself only has an indirect effect (through higher expenditure or budget) on the chance to be promoted (or maintaining one's rich position) - a probability that otherwise depends on the individual ability  $r_i$  and social-capital  $s_i$ . All rich participants and all poor participants having the same behaviour also means that the individual attributes  $r_i$  and  $s_i$  - which are not known to the participant - are not used to decide upon the expenditure (effort) towards the status game.

With  $r_i$  and  $s_i$  uniformly distributed, the likelihood of any participant  $i \in [1, N]$  can be formulated as the probability for their total score  $\mu r_i + (1 - \mu)s_i$  being higher than any other participant - where  $\mu \in (0, 1)$  is a parameter that combines social-capital  $s$  and raw-talent  $r$  so that the participant with maximum score  $\mu r + (1 - \mu)s$  wins the competition. The winner of the competition acquires a rich position so that she obtains the disposable income  $y_2$  while the loser acquires the poor position receive  $y_1 < y_2$ .

To start with a simpler case, consider that in a competition with raw-talents alone, the winning participant is one who has the maximum  $r_i$  drawn from the sample  $N$ . In other words, the winner in a competition of raw-talents is a player  $j$  such that  $\{r_i < r_j \forall i \in [1, N]\}$  and the probability that a given participant  $i$  of raw-skill  $r_i$  is the winner is essentially that of all other players  $j \neq i$  having ability  $r_{j \neq i}$  less than  $r_i$ . This is the probability

$$\int_0^1 P(r_i > r_1) \times P(r_i > r_2) \times \dots P(r_i > r_{i-1}) \times P(r_i > r_{i+1}) \dots P(r > r_N) dr_i$$

With a uniform distribution of skills  $r \in [0, 1]$ , we have  $P(r_i \geq r) = r_i$  for all  $i \in [1, N]$  and therefore, the probability is evaluates to  $\int_0^1 (r_i)^{N-1} dr_i = \frac{1}{N}$ . With more players, the chances of a particular participant winning evidently declines.

Introducing the skills  $s_i$  which are curated in social institutions, we consider a threshold investment  $\bar{v}$  which is necessary to be spent for the social capital to be available to a participant. While this is an arrangement that evidently favours the rich participant, there are also diminishing returns to investing  $\bar{v}$  towards the richer position if naturally occurring raw talent  $r_i$  is also significant in the competition. To direct our focus on conditions when social capital is more important than raw-talent, it assumed in the foregoing analysis that raw-talent is less valued than skills cultivated by institutions<sup>1</sup>. As a result, the condition  $1 - \mu > \mu$  (which is equivalent to the following assumption) has been used throughout in the rest of the discussion.

**Assumption 1.**  $\mu < \frac{1}{2}$

With the threshold investment  $\bar{v}$  as a discrete response to costs of participation, the participants face different probabilities to win depending on whether they spend  $\bar{v}$  or not. More specifically, the participant takes advantage of the skills with weight  $1 - \mu$  only when she spends an amount  $\bar{v}$ . In the all-or-nothing payment scheme, the participant might avoid the investment altogether and keep the investment towards savings instead. The behaviour of all participants depends only on  $v_1, v_2$  - as they are unaware of  $r_i, s_i$  before they participate.

The result depends on the following distribution function of  $x_i \equiv \mu r_i + (1 - \mu)s_i$  under the Assumption 1 where  $r_i$  and  $s_i$  are independent random variables uniformly distributed between 0 and 1 (see Appendix A) .

$$F_X(x) = P(x_k < x) = P(r_k \mu + s_k(1 - \mu) < x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{x-\mu/2}{1-\mu} & \mu < x < 1-\mu \\ 1 - \frac{(x-1)^2}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

In the game where a participant  $i$  with the maximum score  $x_i$  is the winner, the number of rich positions is fixed to  $N - P$  so that only the top-scorer gets to occupy the only rich position available to be filled in every competition. The expected behaviour of the rich and the poor participants in the long-term can be explained with four different cases.

- Case I - The poor participant does not participate due to insufficient investment  $v_1 = 0$  while the rich participant does participate by spending  $v_2 = \bar{v}$ .
- Case II - The rich participant does not participate with  $v_2 = 0$  but the poor participant does participate i.e.  $v_1 = \bar{v}$ .
- Case III - Both poor and rich participants do not participate so that  $v_1 = v_2 = 0$ .
- Case IV - Both poor rich and participant participate with  $v_1 = v_2 = \bar{v}$ .

With Case I, the poor participant can win when her raw-talent (weighted with  $\mu$ ) is higher than the total score of all other poor participants as well as the total score of the rich participants (who compete with acquired skills). The poor are priced out of accessing any skills  $s$  in their score

<sup>1</sup>While effects of changes in  $\mu$  over time on the dynamics of the game are of interest in their own right, the paper narrows our interest on a situations where social capital is weighted higher than raw talent.

in Case I so that only the rich participants with investment  $v_2 = \bar{v}$  have skills  $s > 0$ . Thus, a given poor participant must be ahead both of all of the poor participants and the rich participants so that a participant  $i$  may win only if  $\mu r_i$  is higher than than raw-talent  $\mu r_k$  in all poor participants and also exceeds the score  $\mu r_k + (1 - \mu)s_k$  involving skills among the rich participants

$$P_1(\text{win}|r_i) = \prod_{k=1, k \neq i}^P P(\mu r_k < \mu r_i) \prod_{k=P+1}^N P(\mu r_k + (1 - \mu)s_k < \mu r_i) = r_i^{P-1} F_X(\mu r_i)^{N-P} \quad (1)$$

Therefore,

$$P_1(\text{win}|r_i) = \begin{cases} \frac{\mu^{N-P} r_i^{2N-P-1}}{2\mu(1-\mu)} & 0 < r_i < \mu \\ r_i^{P-1} \left( \frac{\mu r_i - \mu/2}{1-\mu} \right)^{N-P} & \mu < r_i < 1-\mu \\ r_i^{P-1} \left( 1 - \frac{(\mu r_i - 1)^2}{2\mu(1-\mu)} \right)^{N-P} & 1-\mu < r_i < 1 \\ 0 & \text{otherwise} \end{cases}$$

With the uniform density for  $r_i$  (i.e.  $f(r_i) = 1$ ), total probability to win can be calculated as  $\int_{r_i=0}^1 P(\text{win}|r_i) dr_i$ . Since the participants do not know the values of  $r_i, s_i$  before they participate in such competitions, the total probability can be evaluated with the assumption that both  $r_i$  and  $s_i$  range from 0 to 1. Notice also that since only one participant can win the skills-game, the probability  $\int_0^1 \int_0^1 P_2(\text{win}|r_i, s_i) ds_i dr_i$  for a rich participant to win is simply  $\frac{1}{N-P} - \frac{P}{N-P} \int_0^1 \int_0^1 P_1(\text{win}|r_i, s_i) ds_i dr_i$ .

In Case II, when the rich person has no access to social capital (i.e. all rich participants have  $v_2 = 0$ ), the poor participant has the skills-advantage in the game so the probability to win for the poor participant  $i$  is

$$\begin{aligned} P(\text{win}|r_i, s_i) &= \prod_{k=1, k \neq i}^P P(\mu r_k + (1 - \mu)s_k < \mu r_i + (1 - \mu)s_i) \prod_{k=P+1}^N P(\mu r_k < \mu r_i + (1 - \mu)s_i) \\ &= F_X(x_i)^{P-1} \prod_{k=P+1}^N P(\mu r_k < \mu r_i + (1 - \mu)s_i) \quad (2) \end{aligned}$$

In Case III, when neither the rich nor the poor participate in the game, then the game depends only on raw-talents and all participants have the same probability  $\frac{1}{N}$  to win the game. This is because  $\int_{r_i=0}^1 P(\text{win}|r_i) dr_i = \frac{r_i^N}{N} \Big|_0^1 = \frac{1}{N}$  where  $P(\text{win}|r_i) = \prod_{k=1, k \neq i}^P P(r_k < r_i) \prod_{k=P+1}^N P(r_k < r_i) = r_i^{N-1}$ .

In Case IV, when both rich and poor participants set  $v_1 = v_2 = \bar{v}$ , then the poor participant can win only if her score  $x_i$  is above all other participants - rich or poor.

$$\begin{aligned} P(\text{win}|r_i) &= \prod_{k=1, k \neq i}^P P(\mu r_k + (1 - \mu)s_k < \mu r_i + (1 - \mu)s_i) \prod_{k=P+1}^N P(\mu r_k + (1 - \mu)s_k < \mu r_i + (1 - \mu)s_i) \\ &= F_X(x_i)^{N-1} \quad (3) \end{aligned}$$

Using symmetry arguments, this winning probability - which is the same for all participants - also turns out to be  $1/N$  for all the participants.

Notice that with the rich and poor participants having disparate incomes  $y_1$  and  $y_2$  (so that  $y_1 < y_2$ ), the above setup evidently puts the poor participants at a disadvantage since they may not afford to pay the costs to enhance their skills (i.e. realise the true worth of their skills  $s_i$  due to high

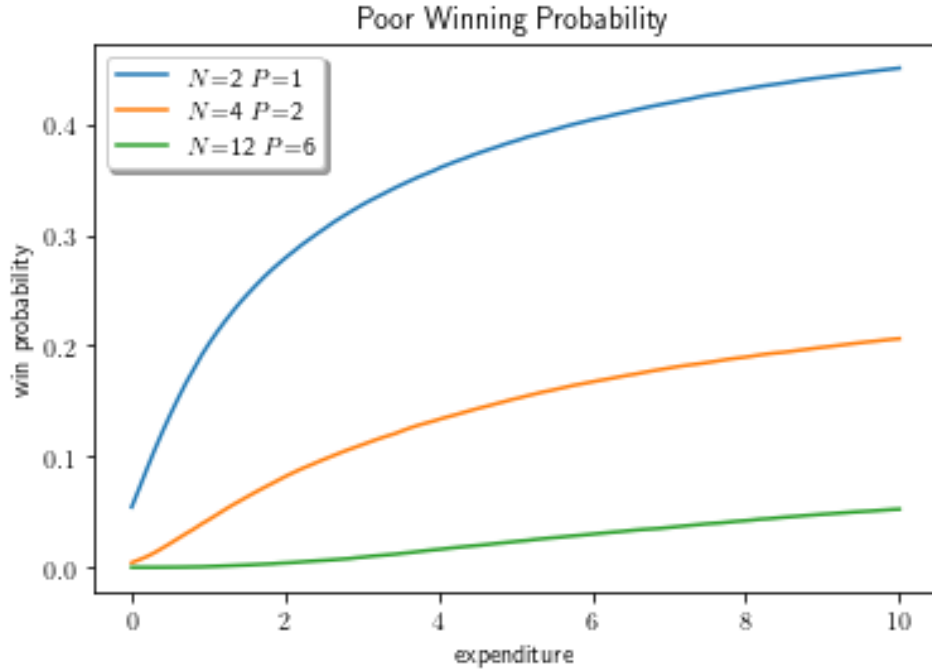


Figure 1: Probability to win for poor participant with increasing number of participants

participation costs). However, since both raw-talent  $r_i$  and social-capital  $s_i$  (the latter enhanced by costly institutional support) are randomly distributed in the population, the competition may never rule out the chance for a poor participant to win. For the same reason, the returns from investments to enhance one's skills may also be diminishing since one cannot beat raw-talent and non-enhanced skills after a certain point.

It is worth emphasising also that the effect of expenditure on the winning probability is weakened with higher number of participants in the game where abilities (both skills  $s_i$  relevant for social capital and the raw-talent  $r_i$ ) are uniformly distributed. In other words, if poor and rich participants were to be considered as two separate groups, then the effect of expenditure on their wins declines when more and more participants are sampled with varied raw-talents and skills. Figures 1 and 2 demonstrate this effect for probability calculated with Monte Carlo simulations for increasing values of the number of poor participants ( $P$ ) and the total number of participants ( $N$ ) while hold the ratio  $P/N$  constant ( $P/N = 1/2$ ). Our Monte Carlo simulations suggest that the role of expenditure is insignificant for  $N \geq 30$ . The effect of income differences using using closed-form solution for  $N = 2$  and  $P = 1$  are now explored in Section 3.1. Equivalent results can be derived using Monte-Carlo simulation for higher values of  $N$  and  $P$ .

### 3.1 Equilibrium Conditions

The solution follows by setting total number of participants  $N = 2$  and the number of poor participants  $P = 1$  in Equations 1, 2 and 3 above. As before, a participant takes advantage of the skills with weight  $1 - \mu$  only by spending an amount  $\bar{v}$  and can decide to avoid the expenditure altogether in order to invest it towards the savings instead. The different probabilities in Case I, II, III and IV are discussed before presenting the role of an additive intertemporal diminishing utility from accumulated wealth. Recall that the behaviour of both participants depends only on  $v_1, v_2$  - as they are unaware of  $r_i, s_i$  before they participate.

An important result for the closed-form solution is the probability distribution (CDF) of  $x_i \equiv \mu r_i + (1 - \mu)s_i$  under assumption  $\mu < 1 - \mu$  (for independent variables  $r_i$  and  $s_i$  that are uniformly distributed between 0 and 1) that is detailed in Appendix A.



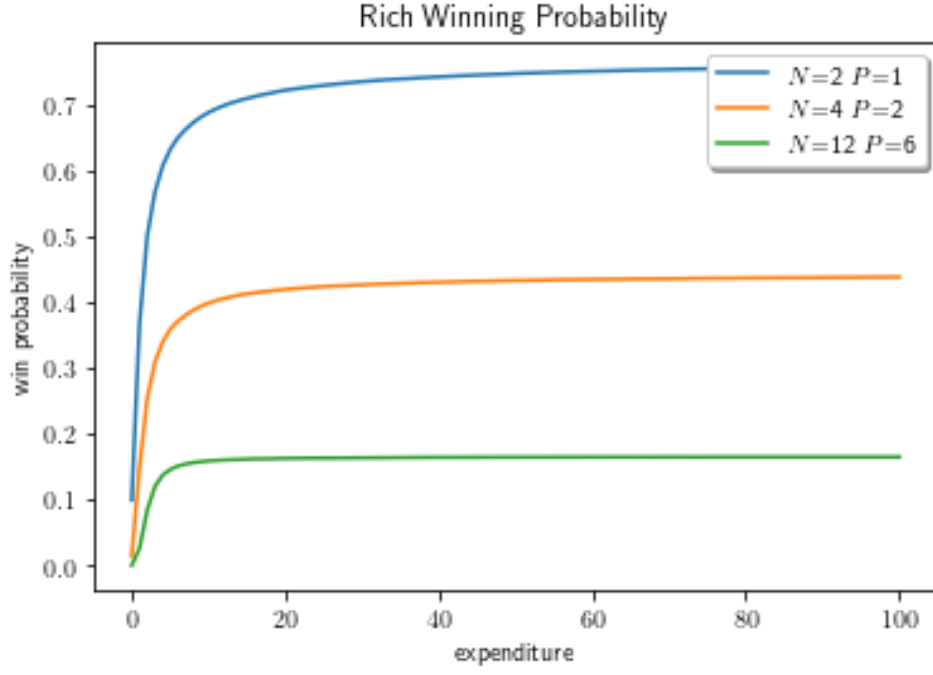


Figure 2: Probability to win for rich participant with increasing number of participants

$$F_X(x) = P(x_k < x) = P(r_k \mu + s_k(1-\mu) < x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{x-\mu/2}{1-\mu} & \mu < x < 1-\mu \\ 1 - \frac{(x-1)^2}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Above probabilities under Case I, II, III and IV are now discussed further.

### Case I

In Case I, the poor participant can win when her raw-talent  $r_1$  is higher than of the total score of rich participant (who competes with acquired skills). The poor participant is priced out of accessing any skills  $s$  in her score so that the skills  $s > 0$  is only true for the rich participant having  $r_2 = \bar{r}$ . Thus, the poor participant with raw-talent  $r_1$  can win with the following probability

$$P(\text{win}|r_1) = P(\mu r_2 + (1-\mu)s_2 < \mu r_1) = F_X(\mu r_1)$$

Notice that  $\mu r_1 < \mu < 1-\mu < 1$  is implied due to  $\mu < \frac{1}{2}$  (and  $r_1 \in (0, 1)$ ). The probability that the total score  $\mu r_2 + (1-\mu)s_2$  does not exceed  $\mu r_1$  implies that only the first part of the piece-wise function for  $F_X(x)$  needs to be considered. Therefore, using  $P(\text{win}|r_1) = \frac{\mu^2 r_1^2}{2\mu(1-\mu)}$ , we evaluate  $P_1(\text{win})$  as

$$P_1(\text{win}) = \int_0^1 P(\text{win}|r_1) dr_1 = \frac{\mu}{6(1-\mu)}$$

Since only one of the participants can win, the probability  $P_2(\text{win})$  for the rich participant to win is simply  $1 - P_1(\text{win})$ . The following thus applies to all other cases (i.e. Cases II, III and IV).

$$P_2(\text{win}) = 1 - P_1(\text{win}) \quad (4)$$

## Case II

In Case II, the rich person does not participate ( $r_2 = 0$ ) and both the poor participants have the skills-advantage in the game. A poor participant wins by having the total score higher than the raw-talent of the rich participant (who does not acquire skills with investment). Representing  $x_i \equiv \mu r_i + (1 - \mu)s_i$ , we have the following probability for the poor participant to win

$$P(\text{win}|r_1, s_1) = P(\mu r_2 < \mu r_1 + (1 - \mu)s_1)$$

Notice that  $P(\mu r_2 < \mu r_1 + (1 - \mu)s_1)$  (i.e.  $P(r_2 < r_1 + (\frac{1}{\mu} - 1)s_1)$ ) is 1 after  $r_1 + (\frac{1}{\mu} - 1)s_1 > 1$  (or  $\mu r_1 + (1 - \mu)s_1 > \mu$ ) and  $\frac{1}{\mu}$  in the interval  $(0, \mu)$  (this is because  $r_2 \in (0, 1)$ ). In other words,

$$P(r_2 < r_1 + (1/\mu - 1)s_1) = \begin{cases} \frac{x_1}{\mu} & 0 < x_1 < \mu \\ 1 & \text{otherwise} \end{cases}$$

The probability to win for the poor participant can be written as (see Appendix A)

$$P(\text{win}|r_1, s_1) = \begin{cases} \frac{x_1}{\mu} & 0 < x < \mu \\ 1 & \mu < x_1 < 1 - \mu \\ 1 & 1 - \mu < x_1 < 1 \\ 0 & \text{otherwise} \end{cases}$$

Using double-integral properties (see Appendix B), the total probability to win  $P_1(\text{win})$  for a poor participant is evaluated as follows

$$\begin{aligned} P_1(\text{win}) &= \int_{s_1=0}^{s_1=1} \int_{r_1=0}^{r_1=1} P(\text{win}|r_1, s_1) d s_1 d r_1 = \int_{r_1=0}^{r_1=1} \int_{s_1=0}^{s_1=\frac{\mu(1-r_1)}{1-\mu}} \frac{(\mu r_1 + (1 - \mu)s_1)}{\mu} d s_1 d r_1 \\ &\quad + \int_{r_1=0}^{r_1=1} \int_{s_1=\frac{\mu(1-r_1)}{1-\mu}}^{s_1=1-\frac{\mu r_1}{1-\mu}} 1 d s_1 d r_1 \\ &\quad + \int_{r_1=0}^{r_1=1} \int_{s_1=1-\frac{\mu r_1}{1-\mu}}^{s_1=1} 1 d s_1 d r_1 \\ &= \frac{\mu}{3(1-\mu)} + \frac{1-2\mu}{1-\mu} + \frac{\mu}{2(1-\mu)} \\ &= \frac{2\mu}{6(1-\mu)} + \frac{6(1-2\mu)}{6(1-\mu)} + \frac{3\mu}{6(1-\mu)} = \frac{6-7\mu}{6(1-\mu)} \end{aligned}$$

The probability  $P_2(\text{win})$  for the rich participant to win follows from Equation 4.

## Case III

In Case III, when neither the rich nor the poor participate in the game, then the game depends only on raw-talents and both participants have the same probability  $\frac{1}{2}$  to win the game. This is because  $\int_{r_i=0}^{r_i=1} P(\text{win}|r_i) d r_i = \frac{r_i^2}{2} \Big|_0^1 = \frac{1}{2}$ . Both the poor and rich participant have the same total probability  $\frac{1}{2}$  to win.

#### Case IV

In Case IV, when both rich and poor participants set  $v_1 = v_2 = \bar{v}$ , then the poor participant can win only if her score  $x_i$  is above that of the rich participant. Therefore,

$$P(\text{win}|r_1, s_1) = P(\mu r_2 + (1-\mu)s_2 < \mu r_1 + (1-\mu)s_1) = F_X(x_1)$$

$$P(\text{win}|r_1, s_1) = \begin{cases} \frac{x_1^2}{2\mu(1-\mu)} & 0 < x_1 < \mu \\ \frac{x_1 - \mu/2}{1-\mu} & \mu < x_1 < 1-\mu \\ 1 - \frac{(x_1-1)^2}{2\mu(1-\mu)} & 1-\mu < x_1 < 1 \\ 0 & \text{otherwise} \end{cases}$$

While this integral can be evaluated based on symmetry arguments (since  $x_i$  is also randomly distributed), the integral  $\int_{s_1=0}^{s_1=1} \int_{r_1=0}^{r_1=1} P(\text{win}|r_1, s_1) dr_1 ds_1$  also evaluates to  $\frac{1}{2}$  (see Appendix B)

$$\begin{aligned} \int_{s=0}^{s=1} \int_{r=0}^{r=1} F_X(x_1) ds_1 dr_1 &= \int_{r_1=0}^{r_1=1} \int_{s_1=0}^{s_1=\frac{\mu(1-r)}{1-\mu}} \frac{(\mu r_1 + (1-\mu)s_1)^2}{2\mu(1-\mu)} ds_1 dr_1 \\ &+ \int_{r_1=0}^{r_1=1} \int_{s_1=\frac{\mu(1-r)}{1-\mu}}^{s_1=1-\frac{\mu r}{1-\mu}} \left( \frac{\mu r_1 + (1-\mu)s_1 - \mu/2}{1-\mu} \right) ds_1 dr_1 \\ &+ \int_{r_1=0}^{r_1=1} \int_{s_1=1-\frac{\mu r}{1-\mu}}^{s_1=1} \left( 1 - \frac{(\mu r_1 + (1-\mu)s_1 - 1)^2}{2\mu(1-\mu)} \right) ds_1 dr_1 \\ &= \frac{\mu^2}{8(1-\mu)^2} + \frac{1-2\mu}{2(1-\mu)} + \frac{(4-5\mu)\mu}{8(1-\mu)^2} = \frac{4(1-\mu)^2}{8(1-\mu)^2} = \frac{1}{2} \end{aligned}$$

As is clear from Figure 3 and above results (see summaries in Tables 3 and 4), the condition  $\mu < 1/2$  - indicating that curated skills are given more importance than raw-talent in the competitions - means that the poor participant is strictly better off (than facing probability 1/2) if she participates while the rich participant doesn't participate (Case II). Also, the poor participant is better off participating for  $\mu < \frac{1}{2}$  when the rich participant does participate as it raises the probability to 1/2. That the competition becomes uncompetitive both when threshold  $\bar{v}$  falls too much (letting both participants participate) or rises too much (letting both participants withdraw) also follows from the properties of the winning probability.

Notice that the rich participant could withdraw and face either 1/2 or lower probability (when the poor participant participates). By participating, therefore, the rich participant would either face probability 1/2 (when poor participant also participates) or better (if the poor participant withdraws). Considering the probabilities alone, the rich participant would always participate. If the rich participant always participates, then so long as budget constraints permit, the poor participant is also better off participating since not participating brings her probability to win below 1/2. If the utility was from probability alone, then both participants would participate and face 1/2 probability.

However, it's not the probability to win that the participants optimise (maximise). A more accurate representation of the real-world is a game where participants face a finite life-time and utility from accumulated assets instead. Thus, what follows is the case where the participants optimise an intertemporally additive utility over their life-time. The life-time in the model is represented with two time-horizons - where the second-time horizon is one before the end of

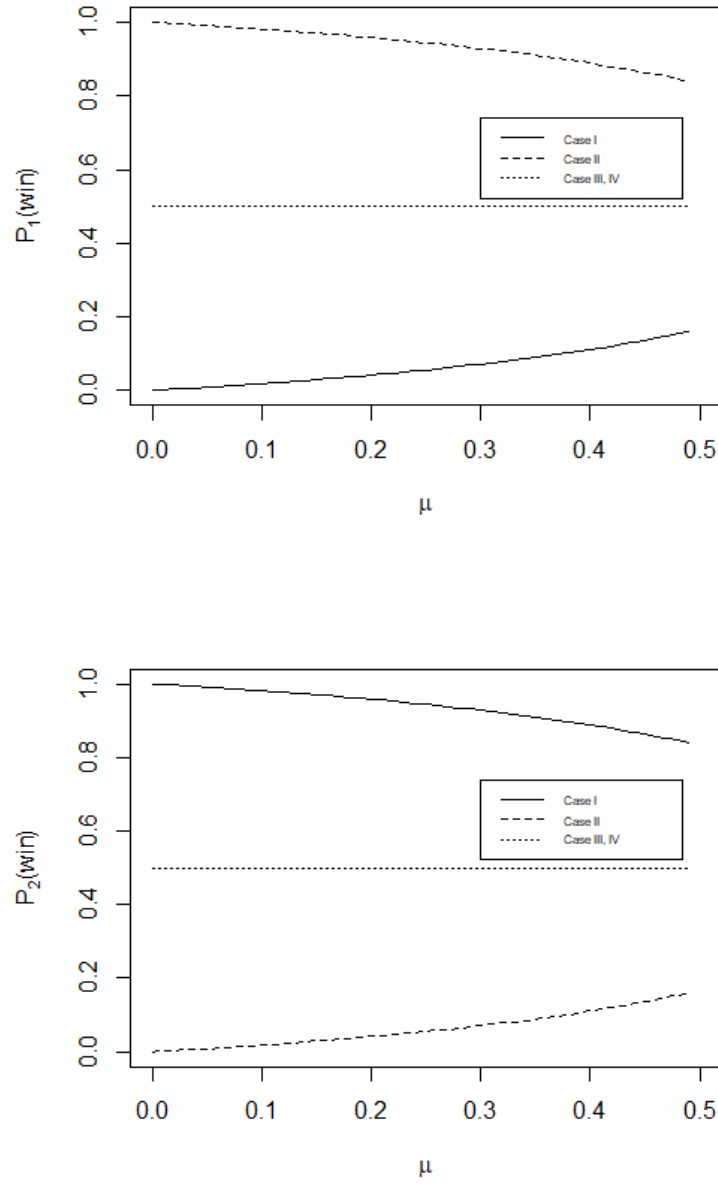


Figure 3: Probability of winning  $P_1(\text{win})$ ,  $P_2(\text{win})$  for poor and rich participants with varying  $\mu$

life when all wealth accumulated by either participant must disappear. With no accumulations at the end of life, both rich and poor participants must decide to spend all their incomes  $y_1$  or  $y_2$  (for poor and rich participants respectively so that  $y_1 < y_2$ ) towards the promotion game. In the first-stage, however, the participants decide the amount to be spent towards promotion based upon the utility from wealth they obtain across the two time-horizons that span their lifetime. For the poor participant, the total utility to be optimised (assuming EUT) under the intertemporally additive utility  $u(\cdot)$ , a winning probability function  $W(x)$  (for expenditure  $x$ ) and a discount-factor  $\delta$  is

$$u(y_1 - v_1) + \frac{1}{\delta}((1 - W(y_1))u(y_1) + W(y_1)u(y_2))$$

Similarly, for the rich participant, the total utility to be optimised (assuming EUT) is

$$u(y_2 - v_2) + \frac{1}{\delta}((1 - W(y_2))u(y_1) + W(y_2)u(y_2))$$

|      |                     |                          |                                    |
|------|---------------------|--------------------------|------------------------------------|
|      |                     | Poor<br><i>withdraws</i> | participant<br><i>participates</i> |
| Rich | <i>withdraws</i>    | Case III                 | Case II                            |
|      | <i>participates</i> | Case I                   | Case IV                            |

Table 1: Cases I,II, III, IV as participant strategies

|                     |                     |                                                                       |                                                                                 |
|---------------------|---------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------|
|                     |                     | Poor                                                                  | participant                                                                     |
|                     |                     | <i>withdraws</i>                                                      | <i>participates</i>                                                             |
| Rich<br>participant | <i>withdraws</i>    | $u(y_2) + \frac{(1 - \frac{P}{N})u(y_2) + \frac{P}{N}u(y_1)}{\delta}$ | $u(y_2) + \frac{u(y_2)p'_2 + (1 - p'_2)u(y_1)}{\delta}$                         |
|                     | <i>participates</i> | $u(y_1) + \frac{(1 - \frac{P}{N})u(y_2) + \frac{P}{N}u(y_1)}{\delta}$ | $u(y_1 - \bar{v}) + \frac{u(y_2)p'_1 + (1 - p'_1)u(y_1)}{\delta}$               |
|                     |                     | $u(y_2 - \bar{v}) + \frac{p_2 u(y_2) + (1 - p_2)u(y_1)}{\delta}$      | $u(y_2 - \bar{v}) + \frac{(1 - \frac{P}{N})u(y_2) + \frac{P}{N}u(y_1)}{\delta}$ |
|                     |                     | $u(y_1) + \frac{p_1 u(y_2) + (1 - p_1)u(y_1)}{\delta}$                | $u(y_1 - \bar{v}) + \frac{(1 - \frac{P}{N})u(y_2) + \frac{P}{N}u(y_1)}{\delta}$ |

Table 2: Utility from two periods for (rich,poor) participants - at the start of the first-period (with  $P$  poor participants among  $N$  total participants)

With above utilities, the conditions for whether the Case I, II, III, IV can become Nash equilibria (see Table 1) are now discussed. A discussion of the implications for the effect of income differences on expenditure towards status then follows. It is worth remarking that the discussion of the effect of the threshold  $\bar{v}$  and the income differences  $y_2 - y_1$  on participation (expenditure) assumes that the exogenous parameter  $\mu$  is stable with respect to short-term changes in threshold investment  $\bar{v}$  as well as disposable incomes  $y_1, y_2$ .

For ease of notation we define  $p_1, p_2, p'_1$  and  $p'_2$  - which correspond to the **two-player** model discussed above.

$$\begin{aligned} p_1 &\equiv \frac{\mu}{6(1-\mu)} \\ p_2 &\equiv 1 - p_1 \\ p'_1 &\equiv \frac{6-7\mu}{6(1-\mu)} \\ p'_2 &\equiv 1 - p'_1 \end{aligned} \tag{5}$$

The choices for the poor and rich participants using a general intertemporally additive function  $u(A)$  (for assets  $A$ ) are shown in Table 2. As we can verify from Figure 3 that we have  $p_1 < 1/2 < p_2$  and  $p'_1 > 1/2 > p'_2$  for all values of  $\mu < 1/2$ .

Given  $\mu < 1/2$ , we have (6)

$$\begin{aligned} p_1 &< 1/2 < p_2 \\ p'_1 &> 1/2 > p'_2 \end{aligned}$$

Notice the Table 2 represents the condition when the budget is non-binding i.e. the poor participant is able to spend  $\bar{v}$ . i.e.  $\bar{v} < y_1 < y_2$ . When the poor participant faces a binding constraint i.e.  $y_1 < \bar{v} < y_2$ , then the rich participant knows that the best case probability to win for the poor participant is  $\frac{1}{2}$ . The rich participant's decision to participate would depend only on utility from savings and the relative advantage from gain in probability. This is elaborated further in Section 3.1.1.

Since the number of poor participants is likely to be higher than that of rich participants in a typical competition, the unequal number of poor and rich participants seems also worth considering in the model. Setting  $N = 3, P = 2$  in the equations 1, 2 and 3, for example, one can interpret the **three-player** model with the following values of  $p_1, p_2, p'_1$  and  $p'_2$

$$\begin{aligned} p_1 &= \frac{\mu}{8(1-\mu)} \\ p_2 &= 1 - 2p_1 \\ p'_1 &= \frac{19\mu^2 - 40\mu + 20}{40(1-\mu)^2} \\ p'_2 &= 1 - 2p'_1 \end{aligned} \tag{7}$$

| Case   | Description                      | Probability of poor participant to win | Response to $\mu$                                                                                                                                                       |
|--------|----------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I      | Only rich invest in skills       | $\frac{\mu}{6(1-\mu)}$                 | With higher $\mu$ , the participants with raw-talent get rewarded. The probability to win increases with higher $\mu$ for the poorer participant.                       |
| II     | Only poor invest in skills       | $\frac{6-7\mu}{6(1-\mu)}$              | With higher $\mu$ , the poor participants with access to skills get rewarded. Therefore, The probability to win decreases with higher $\mu$ for the poorer participant. |
| III,IV | Neither or Both invest in skills | $\frac{1}{2}$                          | Probabilty to win is independent of $\mu$ .                                                                                                                             |

Table 3: The probability of winning for the poor participant in the 3-player model

| Case   | Description                      | Probability of rich participant to win | Response to $\mu$                                                                                                                                                     |
|--------|----------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I      | Only rich invest in skills       | $1 - \frac{\mu}{6(1-\mu)}$             | With higher $\mu$ , the participants with raw-talent get rewarded. Therefore, the probability to win decreases with higher $\mu$ for the rich participant.            |
| II     | Only poor invest in skills       | $1 - \frac{6-7\mu}{6(1-\mu)}$          | With higher $\mu$ , the poor participants with access to skills get rewarded. Therefore, the probability to win increases with higher $\mu$ for the rich participant. |
| III,IV | Neither or Both invest in skills | $\frac{1}{2}$                          | Probability to win is independent of $\mu$ .                                                                                                                          |

Table 4: The probability of winning for the rich participant in the 3-player model

Comparing these probabilities with those from the 2-player model in Equation 5 (also listed in Tables 3 and 4), it can be seen that the probability to win for the poor participant in the 3-player model is lower with respect to the Case III probability (i.e.  $1/N$ ). Likewise, the probability for the rich participant to win is higher in the 3-player model with respect to the Case III probability ( $1/N$ ). The implications for the unequal number of poor and rich participants are further discussed in Section 3.1.2.

### 3.1.1 Binding constraint

Assuming EUT participants, consider the simpler case of binding constraint  $y_1 < \bar{v} < y_2$  first. With the assumption of a stability in long-term participant behaviour (which itself is a result of observed probability of success), mixed-strategies are not considered relevant in the current set up. Considering pure strategies for participation, the rich participant would thus participate only when the utility from participating would be higher than from not participating i.e.  $u(y_2 - \bar{v}) + \frac{p_2 u(y_2) + (1-p_2)u(y_1)}{\delta} > u(y_2) + \frac{u(y_2) + u(y_1)}{2\delta}$ . This results in the following condition

$$\frac{(p_2 - 1/2)(u(y_2) - u(y_1))}{\delta} > u(y_2) - u(y_2 - \bar{v})$$

With income differences being the same an increase in participation cost  $\bar{v}$  discourages participation since RHS increases with  $\bar{v}$  ( $\because u(y_2 - \bar{v})$  is decreasing in  $\bar{v}$ ) and may fail the above condition. Similarly, with participation cost  $\bar{v}$  being the same, an increase in income-difference  $y_2 - y_1$  is in the rich participant's interest whereas a narrowing of income differences reduces



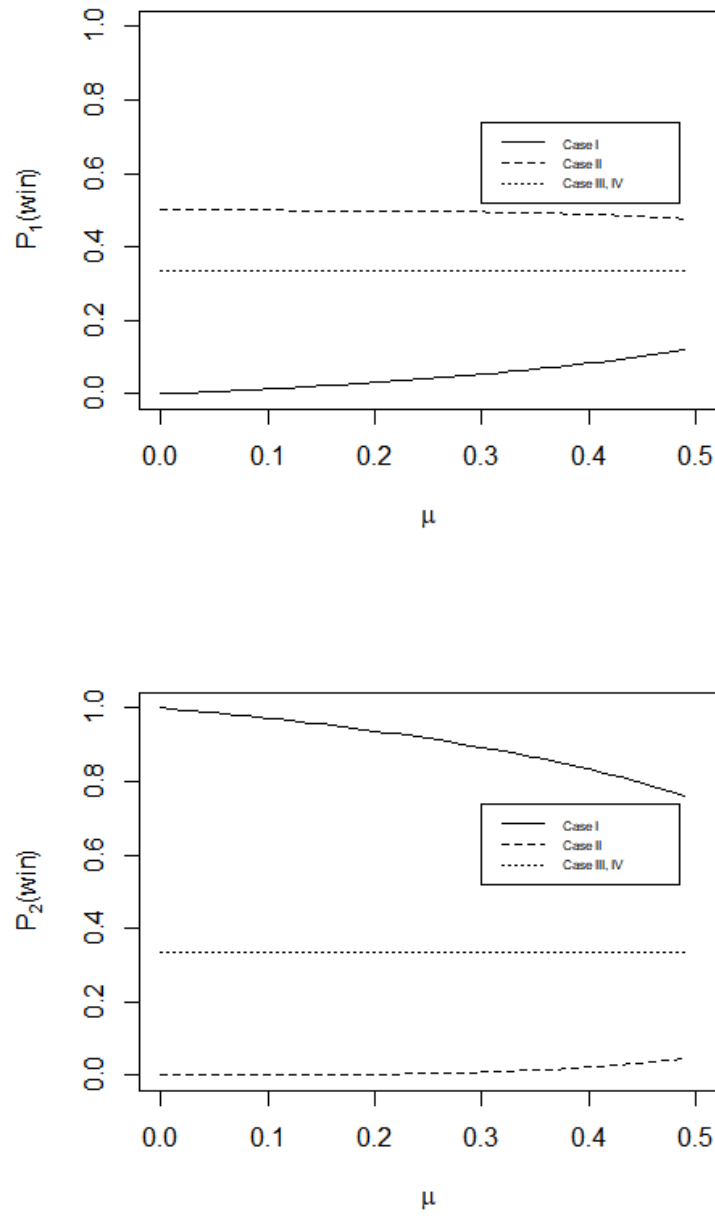


Figure 4: Probability of winning  $P_1(\text{win})$ ,  $P_2(\text{win})$  for poor and rich participants with varying  $\mu$

incentive for the participant to participate by possibly failing the above condition. All else being the same, higher impatience (lower time-discounted value of money or a higher  $\delta$ ) would also discourage participation. Therefore, when budget is binding for the poor participant i.e. if the poor participant cannot participate due to  $y_1 < \bar{v}$ , then narrower income differences, higher participation costs and higher patience for savings (common for both rich and poor participants) all encourage the rich participant to withdraw from the game. This also means that wider income differences ( $y_2 - y_1$ ) combined with lower participation costs ( $\bar{v}$ ) may together help maintain the equilibrium where only the rich participates. Similarly, wider income differences (higher  $y_2 - y_1$ ) and more patience for savings (higher  $\delta$ ) would together maintain the incentive for the rich participant to participate. These conclusions very much fit with the sociological observations made in the literature about status (Hirsch, 1977; Galbraith, 2001).

### 3.1.2 Non-Binding constraint

When budget is non-binding, then participation of the rich participant is her dominant strategy if the condition is also  $u(y_2) + \frac{u(y_2)p'_2 + (1-p'_2)u(y_1)}{\delta} < u(y_2 - \bar{v}) + \frac{u(y_2) + u(y_1)}{2\delta}$  i.e.

$$\frac{(1/2 - p'_2)(u(y_2) - u(y_1))}{\delta} > u(y_2) - u(y_2 - \bar{v})$$

The inferences on how  $y_2 - y_1$ ,  $\bar{v}$  and  $\delta$  influence the participation (see Section 3.1.1) remain the same for the above condition. To discuss the more general Nash equilibria, consider the conditions for all of the Cases I, II, III and IV to become a Nash equilibrium. For ease of notation we define the following for a given  $\bar{v}$ ,  $y_1$  and  $y_2$

$$U_1 \equiv u(y_1) - u(y_1 - \bar{v})$$

$$U_2 \equiv u(y_2) - u(y_2 - \bar{v})$$

One could view  $U_1$  and  $U_2$  as the immediate gain in wealth utility from withdrawing. If  $u(\cdot)$  represents a diminishing utility, then it is also necessary for  $y_1 < y_2$  that

$$U_2 < U_1$$

Consider now that conditions for Case I to be a Nash Equilibrium. If Case I were to be a Nash equilibrium, then for the rich participant's decision to participate, the poor participant must get more utility from withdrawing than from participating and for poor participant's decision to withdraw, the rich must get more utility from participating than from withdrawing. The following conditions <sup>2</sup> must therefore hold simultaneously

$$\begin{aligned} \frac{(\frac{1}{N} - p_1)(u(y_2) - u(y_1))}{\delta} &< U_1 \\ U_2 &< \frac{(p_2 - 1/N)(u(y_2) - u(y_1))}{\delta} \end{aligned} \quad (8)$$

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<sup>2</sup>  $u(y_1) + \frac{p_1 u(y_2) + (1-p_1)u(y_1)}{\delta} > u(y_1 - \bar{v}) + \frac{\frac{1}{N} u(y_2) + \frac{1}{N} u(y_1)}{\delta} \Rightarrow u(y_1) - u(y_1 - \bar{v}) > \frac{(\frac{1}{N} - p_1)(u(y_2) - u(y_1))}{\delta}$  and  $u(y_2 - \bar{v}) + \frac{p_2 u(y_2) + (1-p_2)u(y_1)}{\delta} > u(y_2) + \frac{\frac{1}{N} u(y_2) + \frac{1}{N} u(y_1)}{\delta} \Rightarrow u(y_2) - u(y_2 - \bar{v}) < \frac{(p_2 - 1/N)(u(y_2) - u(y_1))}{\delta}$

Since  $p_2 - \frac{1}{N} \geq \frac{1}{N} - p_1$  (strict inequality follows when the number of poor participants  $P$  is higher than that for rich participants  $N - P$  e.g., with  $N = 3$  and  $P = 2$ ), this is a fairly general condition for the equilibrium. An increase in  $U_1$  due to higher  $\bar{v}$  does not change the equilibrium conditions so long as the corresponding rise  $U_2$  (due to higher  $\bar{v}$ ) does not affect the second inequality. In other words, a higher  $\bar{v}$  favours the equilibrium (Case I) so long as there is an incentive for the rich to participate. Likewise, a decrease in  $\bar{v}$  would be favourable for the equilibrium (Case I) so long as the poor participants do not also have an incentive to participate.

Similar effects can be seen with income differences  $y_2 - y_1$  (which increase  $u(y_2) - u(y_1)$  as well as  $\frac{U_2}{U_1}$ ) if we rewrite Equation 8 as

$$\frac{(\frac{1}{N} - p_1)}{\delta} < \frac{U_1}{u(y_2) - u(y_1)}$$

$$\frac{U_2}{(u(y_2) - u(y_1))} < \frac{(p_2 - 1/N)}{\delta}$$

Limiting our attention to cases when  $\bar{v} < y_1$ , higher  $y_2$  for fixed  $y_1$  decreases  $\frac{U_2}{u(y_2) - u(y_1)}$  and makes it more likely for the rich to participate while it also decreases  $\frac{U_1}{u(y_2) - u(y_1)}$  by raising the stakes enough for the poor to participate (making it less likely for first inequality to be fulfilled). It is easy to see that higher impatience  $\delta$  encourages both poor and rich to withdraw. As would be evident, the equilibrium conditions corresponding to Case II are far more stringent.

If Case II is a Nash Equilibrium, then for rich's decision to withdraw, the poor participant should get more utility from participating than from withdrawing and for poor participant's decision to participate, the rich participant should get more utility from withdrawing than from participating. This means that the following hold<sup>3</sup> simultaneously,

$$U_1 < \frac{(p'_1 - 1/N)(u(y_2) - u(y_1))}{\delta}$$

$$U_2 > \frac{(1/N - p'_2)(u(y_2) - u(y_1))}{\delta}$$

Since  $\frac{(p'_1 - 1/N)(u(y_2) - u(y_1))}{\delta} < \frac{(1/N - p'_2)(u(y_2) - u(y_1))}{\delta}$ , the above cannot be satisfied so long as we have a diminishing utility which necessitates  $U_2 < U_1$ . Thus an equilibrium with Case II which incentivises the poor to participate but not the rich is possible only with a reference dependent utility with  $U_1 < U_2$ . With  $U_1 < U_2$ , we can write above as

$$\frac{U_1}{u(y_2) - u(y_1)} < \frac{(p'_1 - 1/N)}{\delta} < \frac{(1/N - p'_2)}{\delta} < \frac{U_2}{u(y_2) - u(y_1)}$$

Notice that despite a reference-dependent utility, a higher  $\bar{v}$  or more impatience ( $\delta$ ) may disincentivise the poor from participation just the same (leading to Case III). Similarly, higher income differences (an increase in  $y_2$  for fixed  $y_1$ ) could encourage the rich to participate by decreasing  $\frac{U_2}{u(y_2) - u(y_1)}$  (leading to Case IV). Even with the reference dependent utility, the equilibrium with Case II remains a fragile one.

If Case III were to be a Nash-Equilibrium, then for rich participant's decision to withdraw, poor should get more utility from withdrawing than from participating and for poor

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$$^3 u(y_1 - \bar{v}) + \frac{p'_1 u(y_2) + (1 - p'_1) u(y_1)}{\delta} > u(y_1) + \frac{\frac{1}{N} u(y_2) + \frac{1}{N} u(y_1)}{\delta} \Rightarrow u(y_1) - u(y_1 - \bar{v}) < \frac{(p'_1 - 1/N)(u(y_2) - u(y_1))}{\delta} \text{ and } u(y_2) + \frac{p'_2 u(y_2) + (1 - p'_2) u(y_1)}{\delta} >$$

$$u(y_2 - \bar{v}) + \frac{\frac{1}{N} u(y_2) + \frac{1}{N} u(y_1)}{\delta} \Rightarrow u(y_2) - u(y_2 - \bar{v}) > \frac{(1/N - p'_2)(u(y_2) - u(y_1))}{\delta}$$

participant's decision to withdraw, the rich participant should get more from withdrawing than from participating. The following should thus hold<sup>4</sup>,

$$U_1 > \frac{(p'_1 - \frac{1}{N})(u(y_2) - u(y_1))}{\delta}$$

$$U_2 > \frac{(p_2 - \frac{1}{N})(u(y_2) - u(y_1))}{\delta}$$

Since  $p'_1 - \frac{1}{N} \leq p_2 - \frac{1}{N}$  (strict inequality when number of poor participants is higher), we have  $\frac{(p'_1 - \frac{1}{N})(u(y_2) - u(y_1))}{\delta} < \frac{(p_2 - \frac{1}{N})(u(y_2) - u(y_1))}{\delta}$  and  $U_2 < U_1$ . A higher  $\bar{v}$  pushes both  $U_2$  and  $U_1$  further away from participation but a fall in  $\bar{v}$ , incentivises the rich participant to participate (leading to Case I).

$$\frac{U_1}{u(y_2) - u(y_1)} > \frac{(p'_1 - \frac{1}{N})}{\delta}$$

$$\frac{U_2}{u(y_2) - u(y_1)} > \frac{(p_2 - \frac{1}{N})}{\delta}$$

Higher income differences i.e. higher  $y_2$  for fixed  $y_1$  also decrease  $\frac{U_2}{u(y_2) - u(y_1)}$  and incentivise the rich participant to participate (leading to an equilibrium with Case I). Conversely, if the game is not lucrative for the rich it would automatically be infeasible for the poor i.e. if rich withdraws then poor automatically withdraws. If there is any net utility from playing the game, however, Case IV is the likely equilibrium.

If Case IV were to be a Nash Equilibrium, then for rich's decision to participate, the poor participant should get more from participating than from withdrawing and for poor's decision to participate, the rich participant should get more from participating than from withdrawing. Therefore, the following<sup>5</sup> must hold

$$U_1 < \frac{(\frac{1}{N} - p_1)(u(y_2) - u(y_1))}{\delta}$$

$$U_2 < \frac{(\frac{1}{N} - p'_2)(u(y_2) - u(y_1))}{\delta}$$

With  $\frac{(\frac{1}{N} - p'_2)(u(y_2) - u(y_1))}{\delta} < \frac{(\frac{1}{N} - p_1)(u(y_2) - u(y_1))}{\delta}$  and  $U_2 < U_1$ , this equilibrium rests upon the incentive for the poor participant to participate. So long as the poorer participant finds the competition worthwhile, the rich would also participate.

The above results outline how participation costs influence the investments towards status. For very low  $\bar{v}$ , rich and poor would both participate and for very high  $\bar{v}$ , both would withdraw. For everything in between, the only equilibrium with a diminishing utility from wealth is one where rich participate whereas poor don't.

The above framework also allows us to discuss the implications of inequality for the same total wealth in the economy. As such, a segregation where the poor never participate in the status economy seems less favourable than one where both rich and poor participate in the competitions for objective status. The current chapter explains that the likely outcome of a failure to maintain universal participation in status games (under diminishing utility from

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<sup>4</sup>  $u(y_1) + \frac{\frac{1}{N}u(y_2) + \frac{1}{N}u(y_1)}{\delta} > u(y_1 - \bar{v}) + \frac{p'_1 u(y_2) + (1 - p'_1)u(y_1)}{\delta} \Rightarrow u(y_1) - u(y_1 - \bar{v}) > \frac{(p'_1 - \frac{1}{N})(u(y_2) - u(y_1))}{\delta}$  and  $u(y_2) + \frac{\frac{1}{N}u(y_2) + \frac{1}{N}u(y_1)}{\delta} > u(y_2 - \bar{v}) + \frac{p_2 u(y_2) + (1 - p_2)u(y_1)}{\delta} \Rightarrow u(y_2) - u(y_2 - \bar{v}) > \frac{(p_2 - \frac{1}{N})(u(y_2) - u(y_1))}{\delta}$

<sup>5</sup>  $u(y_1 - \bar{v}) + \frac{\frac{1}{N}u(y_2) + \frac{1}{N}u(y_1)}{\delta} > u(y_1) + \frac{p_1 u(y_2) + (1 - p_1)u(y_1)}{\delta} \Rightarrow u(y_1) - u(y_1 - \bar{v}) < \frac{(\frac{1}{N} - p_1)(u(y_2) - u(y_1))}{\delta}$  and  $u(y_2 - \bar{v}) + \frac{\frac{1}{N}u(y_2) + \frac{1}{N}u(y_1)}{\delta} > u(y_2) + \frac{p'_2 u(y_2) + (1 - p'_2)u(y_1)}{\delta} \Rightarrow u(y_2) - u(y_2 - \bar{v}) < \frac{(\frac{1}{N} - p'_2)(u(y_2) - u(y_1))}{\delta}$

wealth) is that of a segregation where only the rich participate in skills leading to economic success. While a reference-based utility (which may be encouraged with segregation) does seem to permit an equilibrium where only the poor are concerned with status games (with respect to skills-enhancement), the result only suggests that a segregation of status games along with that of incomes is another possible long-term outcome of limited participation.

## 4 Conclusion

This paper develops a model where status is conceptualized as a dynamic position sustained through social capital investments (e.g., education, networks, neighborhood selection, professional affiliation). Using a decision theory framework for uncertainty inherent in status considerations, the paper investigates the role of income gaps on expenditures that carry some status while also having an effect on the income mobility of the participants. The endogenisation of status in this manner allows an understanding of the role of long-term inequality on objective status motivations in a more generic framework.

The model provides three important details pertaining to social capital investments in an environment of income inequalities. The first detail is the role of a reference-based utility – which serves as a caveat to the effects of income-differences on status expenditures. The model shows that contentment at lower levels of wealth could support segregation of competitions at different income levels in the economy. In other words, competitions for objective status could be feasible despite vast income differences so long as the utility depends on local levels of wealth. This also seems in line with many of the findings from subjective contentment data on belief in meritocracy amidst inequalities (J. J. Mijs & Savage, 2020).

The second detail highlighted by the model is the interplay of participation costs (threshold) and income-differences in competitions for objective status. The model explains how a withdrawal from expenditures such as education could be the result of a combined effect of both participation costs and income-differences. More specifically, high participation costs push the equilibrium to one where only rich participate in status-related competitions while high income differences may continue to incentivise the poor to participate. In other words, reducing income-differences without lowering of participation costs in status competitions may have an effect of supporting status-games under long-established inequalities.

Lastly, there are only two long-term states of the poor-rich relationship that exist for competitions for status under a diminishing wealth utility - one where both rich and poor participate in competitions for objective status and another where only rich stand to benefit from such competitions. The particular detail demonstrates how income differences limit the benefit from competitions to the poor - providing a quantitative basis for observations on social limits to growth in the literature (Hirsch, 1977; Galbraith, 1994, 1958). More specifically, while poor may be incentivised with status in a setting where status is exclusive to the rich, the other long-term equilibrium is one where both rich and poor must strive for status.

The view of status as dynamic outcome of social capital investments reveals clear limits on how inequality affects status-seeking through mobility-related expenditures. The role of participation costs and reference-based utility in sustaining competitions despite high levels of inequality suggest in particular that participation costs could be a key target for policies towards an inclusive mobility.

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## Appendices

### A Convolution of Uniform Random Variables

The distribution of a convex combination  $w = \mu r + (1 - \mu)s$  of two random variables  $r, s$  - which are distributed according to densities  $f_R(r)$  and  $f_S(s)$  - can be derived using the convolution of two derived variables  $x = \mu r$  and  $y = (1 - \mu)s$ . If  $r$  and  $s$  are uniformly distributed variables ranging from 0 to 1, then  $x$  ranges from 0 to  $\mu$  and  $y$  ranges from 0 to  $(1 - \mu)$ . The CDF for  $F_X(x)$  and  $F_Y(y)$  are simply  $F_X(x) = \frac{x}{\mu}$ ,  $F_Y(y) = \frac{y}{1 - \mu}$  (respectively). Therefore, the densities for  $x$  and  $y$  are  $f_X(x) = \frac{1}{\mu}$  and  $f_Y(y) = \frac{1}{1 - \mu}$ . The density of  $w$  is simple  $f(w) = \int_{-\infty}^{\infty} f_Y(w - x)f_X(x)dx$ . Since,  $f_X(x)$  is  $1/\mu$  only between 0 and  $\mu$ . We have the following integral to evaluate for the convolution

$$f(w) = \frac{1}{\mu} \int_0^{\mu} f_Y(w - x)dx$$

Since  $f_Y(y)$  is 1 only between 0 and  $1 - \mu$ , we have the condition  $0 < w - x < 1 - \mu \Rightarrow w - (1 - \mu) < x < w$ . Further,  $x$  is distributed only between 0 and  $\mu$ . As shown in Figure 5, the integral would evaluate to the following<sup>6</sup> under the assumption  $\mu < \frac{1}{2}$ .

$$f(w) = \frac{1}{\mu} \int_0^{\mu} f_Y(w - x)dx = \begin{cases} \frac{1}{\mu} \int_0^w \frac{1}{1 - \mu} dx & 0 < w < \mu \\ \frac{1}{\mu} \int_0^{\mu} \frac{1}{1 - \mu} dx & \mu < w < 1 - \mu \\ \frac{1}{\mu} \int_{w - (1 - \mu)}^{\mu} \frac{1}{1 - \mu} dx & 1 - \mu < w < 1 \\ 0 & \text{otherwise} \end{cases}$$

This can be simplified as

$$f(w) = \begin{cases} \frac{w}{\mu(1 - \mu)} & 0 < w < \mu \\ \frac{1}{1 - \mu} & \mu < w < 1 - \mu \\ \frac{1 - w}{\mu(1 - \mu)} & 1 - \mu < w < 1 \\ 0 & \text{otherwise} \end{cases}$$

Figure 6 shows this trapezoidal density with  $\mu$  set to  $\frac{1}{5}$ . The CDF  $F_X(x)$  can be evaluated by integrating the above density as  $P(X < x) = \int_{-\infty}^x f(t)dt$ .

<sup>6</sup>Notice that as a convex combination,  $w$  is distributed only between 0 and 1.

$$F_X(x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{\mu}{2(1-\mu)} + \frac{x-\mu}{1-\mu} & \mu < x < 1-\mu \\ \frac{\mu^2}{2\mu(1-\mu)} + \frac{2\mu(1-2\mu)}{2\mu(1-\mu)} + \frac{\mu^2 - x^2 + 2x - 1}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Simplifying above, we have

$$F_X(x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{x-\mu/2}{1-\mu} & \mu < x < 1-\mu \\ 1 - \frac{(x-1)^2}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

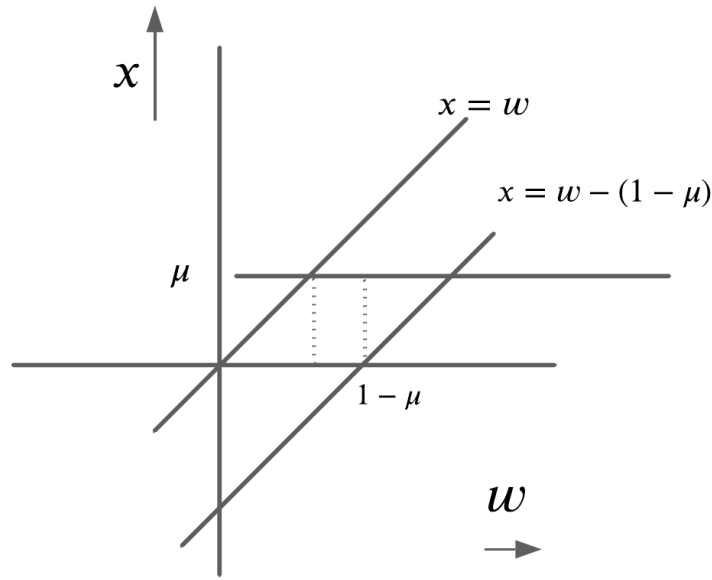


Figure 5: Ranges of  $w$  and  $w - x$  for convolution of two scaled random variables

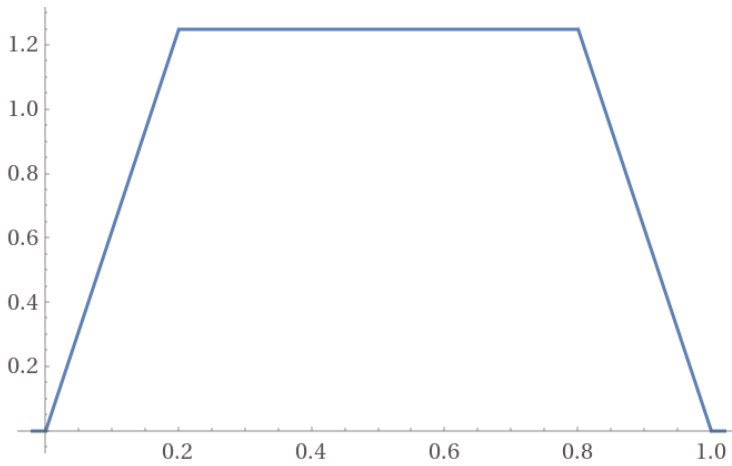


Figure 6: Combined density of  $w$  as the convex combination of two random variables with  $\mu = \frac{1}{5}$



## B $P(\text{win}|r, s)$

To calculate  $\int_{s=0}^{s=1} \int_{r=0}^{r=1} P(\text{win}|r, s) dr ds$  over the variable  $x$  (where  $0 < r < 1$  and  $0 < s < 1$ ), we evaluate the sum of three parts split by the boundaries of the piecewise function. The three parts are shown in Figure 7. If  $x = \mu r + (1 - \mu)s$  and  $F(x)$  is of the following piece-wise form

$$F(x) = \begin{cases} A & 0 < x < \mu \\ B & \mu < x < 1 - \mu \\ C & 1 - \mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Then the integral  $\int_{s=0}^{s=1} \int_{r=0}^{r=1} F(x) dr ds$  can be written as

$$\int_{s=0}^{s=1} \int_{r=0}^{r=1} F(x) ds dr = \int_{r=0}^{r=1} \int_{s=0}^{s=\frac{\mu(1-r)}{1-\mu}} A ds dr + \int_{r=0}^{r=1} \int_{s=\frac{\mu(1-r)}{1-\mu}}^{s=1-\frac{\mu r}{1-\mu}} B ds dr + \int_{r=0}^{r=1} \int_{s=1-\frac{\mu r}{1-\mu}}^{s=1} C ds dr$$

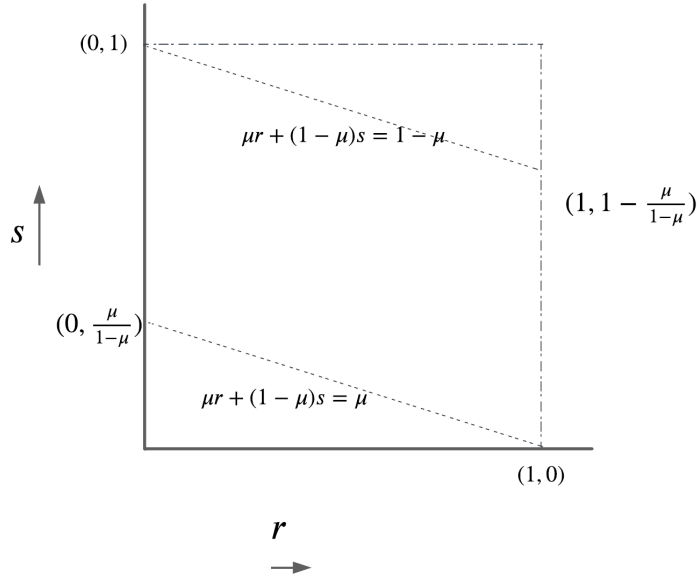


Figure 7: Integration over  $r, s$  split by boundaries  $\mu r + (1 - \mu)s = \mu$  and  $\mu r + (1 - \mu)s = 1 - \mu$